

Econometrics I

Lecture 8a: Panel Data and Fixed Effects

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- ▶ Weeks 5–7: identification by **design** — randomize, or find an instrument.
- ▶ Today: identification by **repeated observation**. We see the same firm, store, customer, or state many times. What does that buy us?
 - Part a (this deck): fixed effects, random effects, and the choice between them.
 - Part b: event studies — the panel version, and the earnings-announcement version.
- ▶ The one idea: things about unit i that *never change* can be removed without ever measuring them.
- ▶ Readings: Hansen Ch. 17; Mixtape Ch. 8.

What a panel is

- ▶ Units $i = 1, \dots, N$ observed at periods $t = 1, \dots, T$ (or T_i if unbalanced).
- ▶ The panels you will meet are **wide and short**: N large, T small.
 - firms $\times$ fiscal years (Compustat: $N \approx 5,000$, $T \approx 10-40$)
 - stores $\times$ weeks (scanner data)
 - customers $\times$ months (subscription, churn)
 - hospitals or physicians $\times$ quarters
- ▶ Everything today is “ $N \rightarrow \infty$, T fixed.” Large- T panels (countries, 50 years) are a different world: unit roots, cointegration, Nickell bias that goes away.
- ▶ Notation: y_{it} outcome, x_{it} a k -vector of regressors. Bars are unit means: $\bar{y}_i = T^{-1} \sum_t y_{it}$.

$$y_{it} = x'_{it}\beta + \alpha_i + \varepsilon_{it}$$

- ▶ α_i : everything about unit i that does not change over t . Management quality, brand, location, a CEO's risk appetite, a customer's taste.
- ▶ ε_{it} : the idiosyncratic shock. Different every period.
- ▶ The conditional mean we want is $\mathbb{E}[y_{it} \mid x_{it}, \alpha_i] = x'_{it}\beta + \alpha_i$. The whole subject: what happens when you *cannot* condition on α_i ?
- ▶ Two names for α_i , and they are not synonyms:
 - a **fixed effect**: treat α_i as a parameter (or a nuisance) — no assumption about how it relates to x_{it} .
 - a **random effect**: treat α_i as a draw, *uncorrelated* with x_{it} .

Running example: management practices and productivity

- ▶ Bloom and Van Reenen (2007) score firms on management practices (monitoring, targets, incentives) and ask whether better practices raise productivity.
- ▶ The obvious worry: well-run firms adopt good practices *and* are productive for other reasons. Firm quality α_i drives both.
- ▶ Our simulated version, with ground truth:

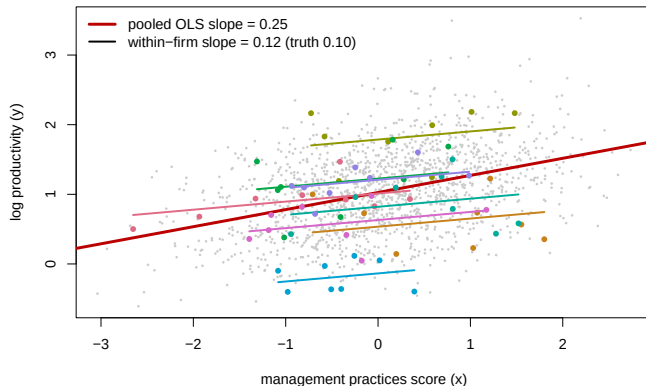
$$y_{it} = 1 + 0.10 x_{it} + \alpha_i + \gamma_t + \varepsilon_{it}, \quad \alpha_i \sim N(0, 0.5^2), \quad \varepsilon_{it} \sim N(0, 0.3^2)$$

$$x_{it} = \rho \alpha_i + (\text{persistent firm component}) + (\text{year-to-year noise}), \quad \rho = 0.7$$

$N = 200$ firms, $T = 8$ years. y is log productivity, x the practices score.

- ▶ We know $\beta = 0.10$. Every estimator gets judged against that number.

The picture



Grey: all firm-years. Colored: eight firms, each drawn with its own intercept and the common within (FE) slope, 0.12 (truth 0.10). The pooled line runs through the *between-firm* cloud.

Pooled OLS: when is it fine?

Ignore the panel structure and regress y_{it} on x_{it} (and a constant).

$$\hat{\beta}_{\text{pooled}} \xrightarrow{p} \beta + \frac{\mathbb{C}(x_{it}, \alpha_i)}{\mathbb{V}(x_{it})}$$

- ▶ Same omitted-variable formula as Week 2. The omitted variable is α_i .
- ▶ Consistent iff $\mathbb{C}(x_{it}, \alpha_i) = 0$ – the “random effects” assumption.
- ▶ In the example, $\mathbb{C}(x, \alpha) > 0$: pooled OLS is biased *up*. Good practices look more effective than they are because good firms have them.
- ▶ Even when unbiased, pooled OLS standard errors are wrong: α_i sits in the error of every period for firm i , so errors are correlated within firm. Cluster by firm (Week 3).

Three ways to get rid of α_i

1. **Dummies (LSDV)**. Put in N firm indicators and run OLS. Fine for $N = 200$; awkward for $N = 5$ million customers.
2. **Within transformation**. Subtract the firm mean from every variable:

$$(y_{it} - \bar{y}_i) = (x_{it} - \bar{x}_i)' \beta + (\varepsilon_{it} - \bar{\varepsilon}_i)$$

$\alpha_i - \bar{\alpha}_i = 0$. It is gone. OLS on the demeaned data is the **fixed-effects (FE) estimator**.

3. **First differences**. $\Delta y_{it} = \Delta x'_{it} \beta + \Delta \varepsilon_{it}$. Also kills α_i .
- ▶ (1) and (2) give *identical* $\hat{\beta}$ (FWL: demeaning is projecting off the dummies). (3) equals them when $T = 2$; differs otherwise.
 - ▶ All three use only **within-firm variation**. That is the price and the point.

The within transformation as a projection

- Stack firm i 's T observations. Let ι be a T -vector of ones and

$$M = I_T - \frac{1}{T}\iota\iota' \quad \text{so that} \quad My_i = y_i - \bar{y}_i\iota.$$

M is symmetric and idempotent: it is the residual-maker from regressing on a constant.

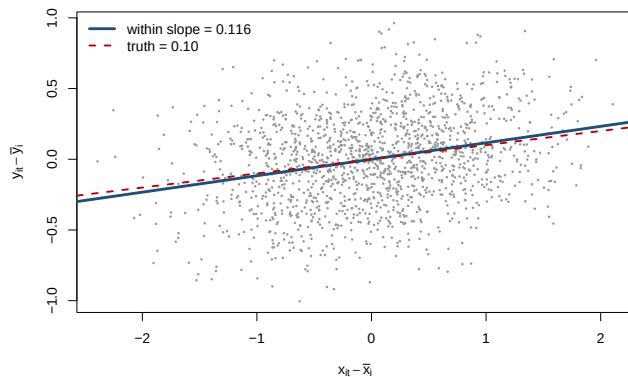
- The FE estimator is OLS after applying M firm by firm:

$$\hat{\beta}_{FE} = \left(\sum_i x_i' M x_i \right)^{-1} \sum_i x_i' M y_i = \left(\sum_i \sum_t \tilde{x}_{it} \tilde{x}_{it}' \right)^{-1} \sum_i \sum_t \tilde{x}_{it} \tilde{y}_{it}$$

with $\tilde{x}_{it} = x_{it} - \bar{x}_i$.

- This is Frisch–Waugh–Lovell from Week 2 with N dummies as the “other” regressors. Software (`fixest`, `pyfixest`) does the demeaning without ever forming the dummies.
- Degrees of freedom: you spent N of them. $\hat{\sigma}_\varepsilon^2$ divides by $NT - N - k$.

The within picture



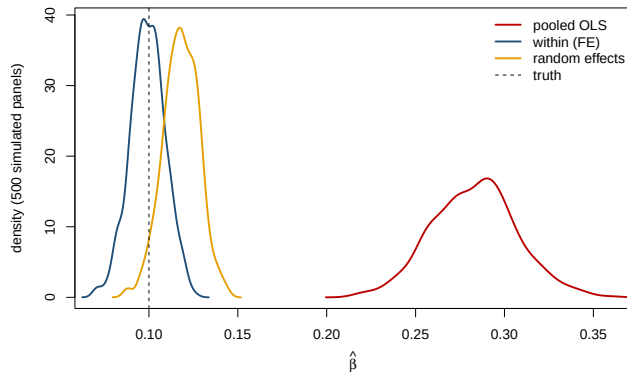
After demeaning, every firm's cloud is centered at the origin. The between-firm variation that carried the bias is gone; what is left is the true slope, plus noise.

Estimates on the simulated panel

	Pooled OLS	First diff.	Within (FE)	Random eff.	Mundlak
$\hat{\beta}$ (truth = 0.10)	0.246	0.105	0.116	0.129	0.116
SE (cluster firm)	(0.023)	(0.013)	(0.011)	(0.011)	(0.011)
Coef. on $\bar{x}_i$					0.376 (0.054)
Firm FE	no	(differenced)	yes	quasi ($\hat{\theta} = 0.74$)	no
$N \times T$	1600	1400	1600	1600	1600

- Pooled OLS is off by a factor of two and a half. FE and first differences are within sampling error of 0.10 (one draw; the next slide shows the distribution).
- Random effects and Mundlak are coming. Note the Mundlak coefficient on $\bar{x}_i$: it is the between-minus-within gap $\hat{\beta}_B - \hat{\beta}_{FE}$ that drives the pooled bias, estimated directly.

Sampling distributions (500 simulated panels)



The pooled estimator is precisely wrong. FE is centered on the truth. RE is in between: it is a weighted average of within and between information, and the between part is contaminated.

Strict exogeneity: the assumption you actually need

For FE to be unbiased we need

$$\mathbb{E}[\varepsilon_{it} \mid x_{i1}, \dots, x_{iT}, \alpha_i] = 0 \quad \text{for every } t.$$

- ▶ Stronger than $\mathbb{E}[\varepsilon_{it} \mid x_{it}] = 0$: today's shock must be unrelated to *past and future* x . Why? Demeaning mixes every period's x into every period's regressor.
- ▶ What it rules out: **feedback**. A firm that has a bad year ($\varepsilon_{it} < 0$) and responds by adopting new practices ($x_{i,t+1} \uparrow$) violates it. A store that sees demand jump and raises prices next week violates it.
- ▶ What it allows: α_i correlated with x in any way at all. That is the whole appeal.
- ▶ First differences need exogeneity only for adjacent periods: weaker than strict exogeneity, but one-period feedback still breaks it. FE and FD differing a lot is itself a diagnostic.

What FE cannot do

- ▶ **Time-invariant regressors drop out.** Industry, founding year, headquarters state: all are absorbed by α_i . FE cannot tell you their effect; it *controls* for them.
- ▶ **Identification comes from movers only.** If practices never change within a firm, $\tilde{x}_{it} = 0$ and the firm contributes nothing. Ask: who changes x , and why?
- ▶ **Measurement error bites harder.** Demeaning removes the persistent (signal) part of x and keeps the noisy part. Attenuation grows:

SD of measurement error in x	0	0.25	0.5	0.75	1.0
Pooled OLS ($\rho = 0$, so unbiased without error)	0.100	0.091	0.080	0.066	0.048
Within (FE)	0.100	0.092	0.071	0.054	0.038

(Here $\rho = 0$, so pooled OLS is unbiased without noise; the point is the comparison.)

- ▶ **Precision.** You are using less variation. Standard errors go up.

Two-way fixed effects

$$y_{it} = x'_{it}\beta + \alpha_i + \gamma_t + \varepsilon_{it}$$

- ▶ γ_t : shocks common to everyone in period t . The business cycle, a platform-wide price change, seasonality, a pandemic.
- ▶ Estimation: demean by unit *and* by period (iterate until both means are zero, which is what `fixest` does). Equivalent to unit dummies plus period dummies.
- ▶ Interpretation of β : the effect of a change in x_{it} *relative to firm i 's usual level and relative to what happened to everyone else in year t .*
- ▶ The workhorse of Weeks 8–9: almost every difference-in-differences you will read is a two-way FE regression.

Random effects: the model as variance components

Suppose α_i is a draw, $\alpha_i \sim (0, \sigma_\alpha^2)$, independent of x and ε . Write the composite error $v_{it} = \alpha_i + \varepsilon_{it}$. Then, for firm i :

$$\mathbb{V}(v_{it}) = \sigma_\alpha^2 + \sigma_\varepsilon^2, \quad \mathbb{C}(v_{it}, v_{is}) = \sigma_\alpha^2 \quad (t \neq s), \quad \text{Corr}(v_{it}, v_{is}) = \frac{\sigma_\alpha^2}{\sigma_\alpha^2 + \sigma_\varepsilon^2} \equiv \rho.$$

- ▶ Every pair of periods within a firm is **equicorrelated** with correlation ρ . Stacked, $\Omega_i = \mathbb{V}(v_i) = \sigma_\varepsilon^2 I_T + \sigma_\alpha^2 \iota \iota'$.
- ▶ Pooled OLS is unbiased under this model but inefficient: it ignores that we have “ N clusters of T correlated observations,” not NT independent ones.
- ▶ The efficient estimator is GLS with this Ω . That is what **random effects** means.

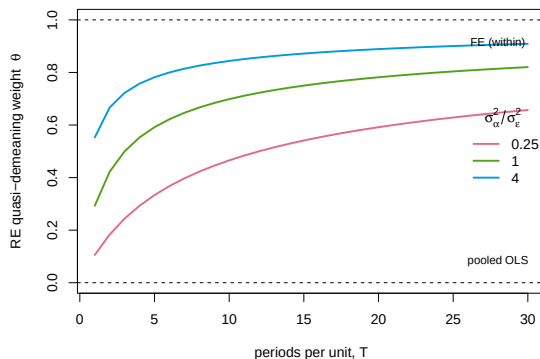
Random effects as quasi-demeaning

GLS with $\Omega_i = \sigma_\varepsilon^2 I_T + \sigma_\alpha^2 \mathbf{u}\mathbf{u}'$ turns out to be OLS on *partially* demeaned data:

$$(y_{it} - \theta \bar{y}_i) = (x_{it} - \theta \bar{x}_i)' \beta + (v_{it} - \theta \bar{v}_i), \quad \theta = 1 - \sqrt{\frac{\sigma_\varepsilon^2}{\sigma_\varepsilon^2 + T\sigma_\alpha^2}}.$$

- ▶ $\theta = 0$: pooled OLS. $\theta = 1$: fixed effects. RE sits in between, and the position is chosen by the data: how much of the variance is α , and how many periods you have.
- ▶ Read the formula: as $T \rightarrow \infty$, $\theta \rightarrow 1$ and RE *becomes* FE. With many periods per unit, the unit mean is estimated so well that GLS just removes it.
- ▶ As $\sigma_\alpha^2 \rightarrow 0$, $\theta \rightarrow 0$: no firm heterogeneity, nothing to remove.
- ▶ Sketch of why: $\Omega_i^{-1/2}$ for an equicorrelated matrix is $\sigma_\varepsilon^{-1} [I_T - \theta \frac{1}{T} \mathbf{u}\mathbf{u}']$, and applying it is exactly the quasi-demeaning above.

How much demeaning? θ as a function of T



For a typical firm panel ($\sigma_\alpha^2 \geq \sigma_\epsilon^2$, $T \approx 10$), θ is already about 0.7 or more. RE and FE will be close in *value* whenever they are both consistent; the question is only whether RE is consistent.

Feasible GLS: estimating the variance components

θ depends on σ_ε^2 and σ_α^2 , which we estimate first (Swamy–Arora):

1. Run the **within** regression. Its residual variance estimates σ_ε^2 (the α_i are gone, so only ε is left):

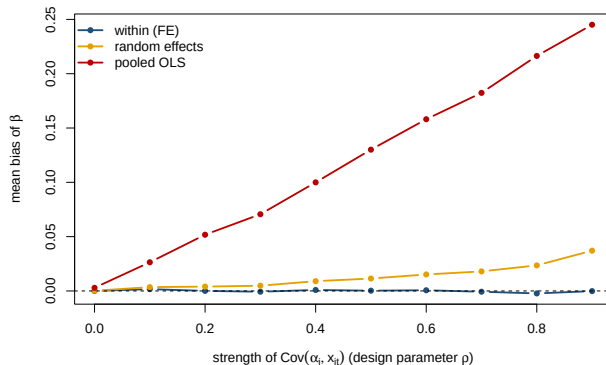
$$\hat{\sigma}_\varepsilon^2 = \frac{\sum_{i,t} \tilde{e}_{it}^2}{NT - N - k}.$$

2. Run the **between** regression: $\bar{y}_i$ on $\bar{x}_i$, one observation per firm. Its residual variance estimates $\sigma_\alpha^2 + \sigma_\varepsilon^2/T$. Subtract: $\hat{\sigma}_\alpha^2 = \hat{\sigma}_{\text{between}}^2 - \hat{\sigma}_\varepsilon^2/T$ (truncate at zero).

3. Form $\hat{\theta}$, quasi-demean, run OLS.

- The within estimator uses only within variation; the between estimator uses only between variation. RE is the precision-weighted combination of the two.
- That sentence is also the diagnosis: if the between variation is contaminated by $\mathbb{C}(\bar{x}_i, \alpha_i) \neq 0$, RE inherits part of the pooled bias (less than $(1 - \theta)$ of it).
- The workbook implements this in a dozen lines of R; `plm` and `lme4` do it for you.

RE bias as the correlation between α_i and x_{it} grows



FE does not care. RE's bias grows with the correlation, less than pooled OLS because θ is large here ($T = 8$), but biased. Nothing about “random” protects you from this.

Fixed vs. random: what the words actually mean

- Both treat α_i as unobserved and ε_{it} as random. The difference is *one assumption*:

$$\text{RE: } \mathbb{E}[\alpha_i \mid x_{i1}, \dots, x_{iT}] = 0$$

$$\text{FE: no restriction on } \mathbb{E}[\alpha_i \mid x_{i1}, \dots, x_{iT}]$$

- Under the RE assumption both are consistent and RE is more efficient; under its failure only FE is consistent. Bias beats variance: economists default to FE. (Hansen: the distinction is about what you assume, not whether α_i is “really” random.)
- Why anyone runs RE anyway:
- you need the coefficient on a **time-invariant** regressor (industry, gender, region);
 - you believe the assumption because x was **randomized** across units;
 - you care about α_i itself and want to **shrink** noisy unit means (teacher effects, hospital quality, seller ratings) — “empirical Bayes” is RE by another name;
 - your field calls it a **hierarchical** or **mixed** model and everyone uses it.

The Hausman test

Under the RE assumption, $\hat{\beta}_{FE}$ and $\hat{\beta}_{RE}$ estimate the same thing; if it fails, they diverge. So test whether they differ:

$$H = (\hat{\beta}_{FE} - \hat{\beta}_{RE})' [\hat{V}_{FE} - \hat{V}_{RE}]^{-1} (\hat{\beta}_{FE} - \hat{\beta}_{RE}) \xrightarrow{d} \chi_k^2 \quad \text{under } H_0.$$

- ▶ The variance of the difference collapses to $V_{FE} - V_{RE}$ because RE is efficient under H_0 (the efficient estimator is uncorrelated with the difference). Neat, but fragile:
 - it requires the *classical* RE variance — no clustering, no heteroskedasticity;
 - $\hat{V}_{FE} - \hat{V}_{RE}$ need not be positive definite in finite samples;
 - only regressors that vary within unit can be compared.
- ▶ Rejecting says “RE is inconsistent, use FE.” Not rejecting says “you could not tell”: with few periods RE and FE are noisy and the test has little power.
- ▶ There is a better way to run the same test.

Mundlak: put $\bar{x}_i$ in the regression

Model the correlation instead of assuming it away. Let

$$\alpha_i = \bar{x}_i' \gamma + u_i, \quad \mathbb{E}[u_i \mid x_{i1}, \dots, x_{iT}] = 0.$$

Substitute:

$$y_{it} = x_{it}' \beta + \bar{x}_i' \gamma + u_i + \varepsilon_{it}.$$

- ▶ Now u_i satisfies the RE assumption by construction. Run RE (or even pooled OLS) on $(x_{it}, \bar{x}_i)$.
- ▶ Two facts (Mundlak 1978; the **correlated random effects** model): the coefficient on x_{it} is *exactly* $\hat{\beta}_{FE}$, and $\hat{\gamma}$ is exactly the between-minus-within difference. The Hausman test is $H_0 : \gamma = 0$.
- ▶ Advantages over the classical Hausman statistic: cluster-robust standard errors are fine; nothing needs to be positive definite; you can keep time-invariant regressors in the model.

Both tests on the simulated panel

Test	Statistic	p -value	Verdict
Hausman $(\hat{\beta}_{FE} - \hat{\beta}_{RE})^2 / (\hat{V}_{FE} - \hat{V}_{RE})$	241.0	< 0.001	reject RE
Mundlak t -test on $\bar{x}_i$ (cluster-robust)	6.9	< 0.001	reject RE

- ▶ The classical test works here because FE and RE are the *same model* (both with year effects) and the errors are homoskedastic by design. It rejects.
- ▶ Get the specification wrong — one-way RE against two-way FE — and $\hat{V}_{FE} - \hat{V}_{RE}$ turns negative in this very sample: the fragility from the previous slide.
- ▶ The Mundlak version agrees, rejects decisively, and came with clustered standard errors for free.
- ▶ Look back at the estimates table: the Mundlak coefficient on x_{it} matched FE to three decimals.

Inference: cluster by the unit

- ▶ After demeaning, the errors $\varepsilon_{it} - \bar{\varepsilon}_i$ are correlated within firm mechanically, and ε_{it} itself is usually serially correlated too.
- ▶ Week 3's answer applies unchanged: cluster at the firm. Asymptotics are in N , the number of clusters, which is why “ N large, T small” is the comfortable case.
- ▶ Bertrand, Duflo and Mullainathan (2004): in state-year panels with serially correlated outcomes, unclustered two-way FE regressions reject a true null of *no effect* up to 45% of the time at the 5% level. Cluster.
- ▶ Cluster at the level of the fixed effect *or* the level at which x is assigned, whichever is coarser. Firm FE with a state-level policy: cluster by state.
- ▶ Few clusters (under ≈ 40) is a separate problem; wild cluster bootstrap (Week 3).

Pitfall: lagged outcomes (Nickell 1981)

$$y_{it} = \rho y_{i,t-1} + x'_{it}\beta + \alpha_i + \varepsilon_{it}$$

- ▶ Tempting: “control for last year’s productivity.” Demean it:

$$(y_{it} - \bar{y}_i) = \rho (y_{i,t-1} - \bar{y}_i) + \dots + (\varepsilon_{it} - \bar{\varepsilon}_i)$$

- ▶ $\bar{y}_i$ contains y_{it} , which contains ε_{it} ; and $\bar{\varepsilon}_i$ contains $\varepsilon_{i,t-1}$, which is in $y_{i,t-1}$. Regressor and error are correlated **by construction**.
- ▶ The bias is $O(1/T)$: it does not vanish as $N \rightarrow \infty$. With $T = 8$ and $\rho = 0.5$, $\hat{\rho}$ is biased down by roughly 0.2, and $\hat{\beta}$ picks up some of the damage.
- ▶ Fixes (Anderson–Hsiao, Arellano–Bond) instrument $\Delta y_{i,t-1}$ with deeper lags. Rule for this course: no lagged dependent variable in an FE regression unless you know why.

Partial pooling: no pooling, complete pooling, and the thing in between

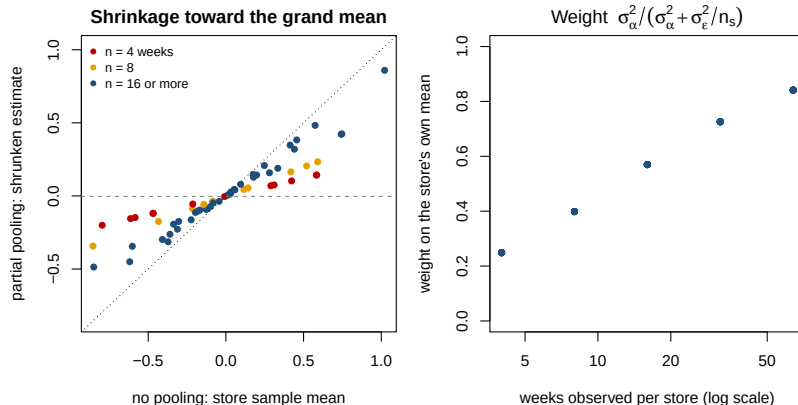
60 stores, 4 to 64 weeks each, true store effects $\alpha_s \sim N(0, 0.4^2)$, weekly noise SD 1. Three estimates of each store's effect:

- ▶ **No pooling:** the store's own mean (fixed effects). Unbiased; noisy when n_s is small.
- ▶ **Complete pooling:** the grand mean for everyone (pooled OLS). Precise; ignores that stores differ.
- ▶ **Partial pooling:** $\hat{\alpha}_s^{PP} = \bar{y} + w_s(\bar{y}_s - \bar{y})$ with

$$w_s = \frac{\sigma_\alpha^2}{\sigma_\alpha^2 + \sigma_\varepsilon^2/n_s}.$$

The random-effects prediction, empirical Bayes, or a hierarchical model with a normal prior: three names, one formula. The weight is the store's reliability: the signal share of its mean's variance.

Partial pooling, pictured



Left: four-week stores (red) are pulled most of the way to the grand mean; 64-week stores keep most of their own mean. Right: the weight by weeks observed. MSE against the true α_s : no pooling 0.082, complete pooling 0.124, partial pooling 0.053.

Why marketing uses this all the time

- ▶ Partial pooling beats both extremes on mean squared error: it is the bias–variance tradeoff from Lecture 1b, applied sixty times with a data-driven weight.
- ▶ The **ranking problem**: sort stores by sample mean and the top five include small- n stores that got lucky; sort by the shrunken estimate and the top five are stores with enough data to be believed. Same logic for sellers, products, sales reps, ad creatives, teachers.
- ▶ Hierarchical Bayes in marketing (Rossi, Allenby and McCulloch 2005) is this with the prior estimated from the data and full posteriors instead of point estimates: customer-level coefficients in a conjoint or choice model are pooled toward the population with weights set by how much each customer's data can support. `lme4::lmer`, `brms` in R; `statsmodels.MixedLM`, `PyMC` in Python.
- ▶ When β is the object and α_i a nuisance: FE. When the α_i themselves are the product: partially pool.

Pitfall: when α_i is the object of interest

- ▶ Sometimes the fixed effect *is* the estimand: which physicians have better outcomes, which managers add value, which sellers deliver on time.
- ▶ $\hat{\alpha}_i = \bar{y}_i - \bar{x}_i' \hat{\beta}$ is unbiased but noisy with variance $\approx \sigma_\varepsilon^2 / T_i$. The units at the top of the ranking are disproportionately the ones with small T_i and lucky draws.
- ▶ Shrink toward the mean in proportion to noise: the partial-pooling weight from the previous slides.
- ▶ Believing $\hat{\alpha}_i$ causally also needs as-good-as-random *assignment* of units to i : patients are not randomly assigned to physicians (movers designs: Chetty–Hendren for places, Finkelstein–Gentzkow–Williams for physicians).
- ▶ Nonlinear models (logit with firm FE) add the **incidental parameters** problem: $\hat{\beta}$ is inconsistent for fixed T .

R (fixest):

```
feols(y ~ x | firm + year, data = d, cluster = ~firm)      # two-way FE
d[, xbar := mean(x), by = firm]
feols(y ~ x + xbar | year, data = d, cluster = ~firm)      # Mundlak / CRE
```

Python (pyfixest):

```
pf.feols("y ~ x | firm + year", data=d, vcov={"CRV1": "firm"})
d["xbar"] = d.groupby("firm")["x"].transform("mean")
pf.feols("y ~ x + xbar | year", data=d, vcov={"CRV1": "firm"})
```

- Random effects proper: `plm(..., model = "random")` or `lme4::lmer(y ~ x + (1|firm))`; `statsmodels.MixedLM` in Python. The workbook writes FGLS by hand so you see $\hat{\theta}$.

- ▶ α_i is everything time-invariant about unit i . Pooled OLS is biased by $\mathbb{C}(x, \alpha) / \mathbb{V}(x)$.
- ▶ FE removes α_i by demeaning (or dummies, or differencing) and needs **strict exogeneity**: no feedback from shocks to future x . It uses within variation only.
- ▶ RE is GLS with equicorrelated errors, i.e. quasi-demeaning with weight θ . It needs $\alpha_i \perp x$, and is biased otherwise.
- ▶ Mundlak: add $\bar{x}_i$. You get FE's slope, a robust Hausman test, and time-invariant regressors, all at once.
- ▶ Cluster by unit. Do not lag the outcome. Two-way FE is the template for next hour and next week.

`inclass_session8.Rmd`:

1. Reproduce the estimates table: pooled, within, FD, hand-rolled RE, Mundlak.
2. Turn ρ down to zero and up to 0.9. Watch RE move and FE stay put.
3. Add measurement error to x . Which estimator suffers more, and why?